

PASSIVE COOLING SYSTEM FOR HOMES

Problem Statement

Homes located in hot and humid regions often experience excessive indoor temperatures due to high solar radiation, poor ventilation, inappropriate building materials, and lack of proper shading, forcing occupants to depend on energy-intensive mechanical cooling systems such as air-conditioners and fans. This leads to increased electricity consumption, higher living costs, and growing environmental impacts. Therefore, there is a need to develop an effective, low-cost, and sustainable passive cooling system for homes that can naturally reduce indoor temperature by optimizing ventilation, shading, thermal mass, evaporative effects, and building orientation. The project aims to design, model, and test such passive strategies to improve indoor comfort without relying heavily on external energy sources.

Introduction

Passive cooling is a sustainable building approach that reduces indoor temperatures using natural processes such as ventilation, shading, thermal mass, evaporative cooling, and radiative heat loss, without relying on energy-intensive mechanical systems. As climate conditions become hotter and electricity demand rises, passive cooling offers an efficient and low-cost alternative for maintaining indoor comfort in residential homes. By optimizing building orientation, materials, window placement, airflow pathways, and solar protection, passive cooling strategies help minimize heat gain and enhance

natural cooling throughout the day. This project explores the principles, design methods, and performance evaluation of passive cooling systems to demonstrate their potential for improving thermal comfort and reducing energy consumption in modern homes.

3. Need of Passive Cooling System for Homes

a. Energy Crisis & Rising Electricity Bills

Air-conditioners account for a large portion of household energy use. Passive cooling reduces dependence on these appliances.

b. Environmental Sustainability

Mechanical cooling increases electricity demand and carbon emissions. Passive systems lower environmental impact and support low-carbon living.

c. Improved Indoor Comfort Without Power

During power cuts or peak summer months, passive cooling ensures homes remain comfortable.

d. Affordable & Accessible

Passive cooling requires simple materials and design changes—ideal for rural, low-income, and sustainable housing.

e. Urban Heat Island Mitigation

By using shading, vegetation, reflective roofs, and ventilation, passive cooling decreases localized heating.

f. Extends Building Lifespan

Better thermal control reduces thermal stress on building materials.

4. Technical Changes Required for the Project

a. Building Envelope Improvements

- High-reflective cool roof coatings
- Insulated or ventilated roof layers
- Use of thermal mass materials (brick, adobe, stone)

b. Ventilation Optimization

- Proper placement of openings based on wind direction
- Cross-ventilation paths
- Stack ventilation using solar chimneys
- Operable vents and clerestories

c. Solar Heat Gain Control

- External shading devices (overhangs, louvers, pergolas)
- Vegetative shading (trees, creepers)
- Jali screens and perforated facades

d. Evaporative Cooling Add-ons

- Wetted pads or clay pots
- Water bodies or rooftop sprinkling
- Courtyard cooling with vegetation

e. Radiative Cooling Enhancements

- High-emissivity roof surfaces
- Night-time flushing (night ventilation)

f. Monitoring & Data Acquisition

- Temperature and humidity sensors (DHT22, DS18B20)
- Anemometer for airflow
- Arduino/ESP32 for logging data

g. Thermal and Airflow Modeling

- Lumped thermal model (3-node or multi-zone)
- Airflow equations (wind-driven + stack effect)
- Solar gain calculations and sky temperature estimation

5. Project Aim

To design, model, and evaluate a passive cooling system for residential homes that reduces indoor temperature naturally by combining strategies such as cross-ventilation, shading, thermal mass, evaporative cooling, and radiative cooling.

The project aims to:

1. Develop a mathematical and simulation model of passive cooling performance.
2. Construct a physical prototype or small-scale model implementing key passive technologies.
3. Measure indoor temperature, ventilation rate, and surface temperatures.

4. Analyze the effectiveness and quantify temperature reduction.
5. Recommend an optimized passive cooling design suitable for real homes.

Conclusion

The study demonstrates that passive cooling systems are a practical, energy-efficient, and environmentally sustainable solution for maintaining comfortable indoor temperatures in residential buildings. By integrating strategies such as enhanced ventilation, effective shading, cool roof technologies, thermal mass utilization, and evaporative or radiative cooling techniques, significant reductions in indoor heat gain can be achieved without mechanical devices. The results highlight that passive design not only improves thermal comfort but also reduces energy consumption, operational costs, and carbon footprint. Overall, passive cooling provides a viable approach for creating climate-responsive, affordable, and sustainable homes suitable for diverse climatic conditions.